

自旋匹配 TDA

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1 $S_f = S_i \geq 1$ 的 S-TDA

根据 Li 和 Liu 文章的 Table I, $S_f = S_i$ 可以分成 $S_i = \frac{1}{2}$ 和 $S_i \geq 1$ 两种情况 ($S_i = 0$ 的情况自动回归闭壳层)。

当 $S_f = S_i \geq 1$ 时, 根据式 (104) 任意矩阵 $M_{pq,rs}$ 为

$$\begin{pmatrix} M_{00} & -\sqrt{\frac{S+1}{S}}M_{01} \\ -\sqrt{\frac{S+1}{S}}M_{10} & -\frac{1}{S}M_+ + \frac{S+1}{S}M_0 \end{pmatrix} \quad (1)$$

其中左上角为单重态激发块, 右下角为三重态激发块。它们的定义是

$$[A_{00}]_{pq,rs} = \langle SS|O_{pq}(0,0)H_N O_{rs}^\dagger(0,0)|SS\rangle, \quad (2)$$

$$[A_{01}]_{pq,rs} = \langle SS|O_{pq}(0,0)H_N O_{rs}^\dagger(1,0)|SS\rangle, \quad (3)$$

$$[A_{10}]_{pq,rs} = \langle SS|O_{pq}(1,0)H_N O_{rs}^\dagger(0,0)|SS\rangle, \quad (4)$$

$$[A_-]_{pq,rs} = \langle SS|O_{pq}(1,-1)H_N O_{rs}^\dagger(1,-1)|SS\rangle, \quad (5)$$

$$[A_0]_{pq,rs} = \langle SS|O_{pq}(1,0)H_N O_{rs}^\dagger(1,0)|SS\rangle. \quad (6)$$

其中 $H_N = H - E_{\text{ref}}$ 为正规化的 Hamiltonian

$$H_N = \sum_{PQ} f_{PQ} N(a_P^\dagger a_Q) + \frac{1}{4} \sum_{PQRS} \langle PQ||RS\rangle N(a_P^\dagger a_Q^\dagger a_S a_R). \quad (7)$$

1.1 三重态激发块的 A 和 S

对于 CV-CV 块, 有

$$[A_+]_{ai,bj} = \langle SS|(a_{i\beta}^\dagger a_{a\alpha})H_N(a_{b\alpha}^\dagger a_{j\beta})|SS\rangle \quad (8)$$

$$= \delta_{ij}f_{ab}^\alpha - \delta_{ab}f_{ji}^\beta - \langle a\alpha j\beta||b\alpha i\beta\rangle \quad (9)$$

$$= \delta_{ij}f_{ab}^\alpha - \delta_{ab}f_{ji}^\beta + \langle a\alpha j\beta||i\beta b\alpha\rangle. \quad (10)$$

$$[A_0]_{ai,bj} = \frac{1}{2} \langle SS|(a_{i\alpha}^\dagger a_{a\alpha} - a_{i\beta}^\dagger a_{a\beta})H_N(a_{b\alpha}^\dagger a_{j\alpha} - a_{b\beta}^\dagger a_{j\beta})|SS\rangle \quad (11)$$

$$= \frac{1}{2}\delta_{ij}(f_{ab}^\alpha + f_{ab}^\beta) - \frac{1}{2}\delta_{ab}(f_{ji}^\alpha + f_{ji}^\beta) + \frac{1}{2}[\langle a\alpha j\alpha||i\alpha b\alpha\rangle + \langle a\alpha j\beta||b\beta i\alpha\rangle + \langle a\beta j\alpha||b\alpha i\beta\rangle + \langle a\beta j\beta||i\beta b\beta\rangle] \quad (12)$$

$$= \frac{1}{2}\delta_{ij}(f_{ab}^\alpha + f_{ab}^\beta) - \frac{1}{2}\delta_{ab}(f_{ji}^\alpha + f_{ji}^\beta) + \frac{1}{2}[\langle a\alpha j\alpha||i\alpha b\alpha\rangle - \langle a\alpha j\beta||i\alpha b\beta\rangle - \langle a\beta j\alpha||i\beta b\alpha\rangle + \langle a\beta j\beta||i\beta b\beta\rangle]. \quad (13)$$

$$[A_-]_{ai,bj} = \langle SS|(a_{i\alpha}^\dagger a_{a\beta})H_N(a_{b\beta}^\dagger a_{j\alpha})|SS\rangle \quad (14)$$

$$= \delta_{ij}f_{ab}^\beta - \delta_{ab}f_{ji}^\alpha + \langle a_\beta j\alpha || i\alpha b\beta \rangle. \quad (15)$$

$$A_{ai,bj} = \frac{1}{2}\delta_{ij}\left[\left(1 - \frac{1}{S}\right)f_{ab}^\alpha + \left(1 + \frac{1}{S}\right)f_{ab}^\beta\right] - \frac{1}{2}\delta_{ab}\left[\left(1 + \frac{1}{S}\right)f_{ji}^\alpha + \left(1 - \frac{1}{S}\right)f_{ji}^\beta\right] + \left(1 + \frac{1}{S}\right)K_{ai,bj}^T - \frac{1}{S}K_{ai,bj}^{\text{SF}}. \quad (16)$$

$$A_{ai,bj}^{1/2} = -\frac{1}{2}\delta_{ij}f_{ab}^\alpha + \frac{3}{2}\delta_{ij}f_{ab}^\beta - \frac{3}{2}\delta_{ab}f_{ji}^\alpha + \frac{1}{2}\delta_{ab}f_{ji}^\beta + \frac{1}{3}K_{ai,bj}^T + \frac{2}{3}K_{ai,bj}^{\text{SF}}. \quad (17)$$

对于 CV-CO 块, 有

$$[A_+]_{ai,vj} = 0. \quad (18)$$

$$[A_0]_{ai,vj} = \frac{1}{2}\langle SS|(a_{i\alpha}^\dagger a_{a\alpha} - a_{i\beta}^\dagger a_{a\beta})H_N(-a_{v\beta}^\dagger a_{j\beta})|SS\rangle \quad (19)$$

$$= \frac{1}{2}\delta_{ij}f_{av}^\beta + \frac{1}{2}[\langle a\alpha j\beta || v\beta i\alpha \rangle + \langle a\beta j\beta || i\beta v\beta \rangle] \quad (20)$$

$$= \frac{1}{2}\delta_{ij}f_{av}^\beta + \frac{1}{2}[-\langle a\alpha j\beta || i\alpha v\beta \rangle + \langle a\beta j\beta || i\beta v\beta \rangle]. \quad (21)$$

$$[A_-]_{ai,vj} = \langle SS|(a_{i\alpha}^\dagger a_{a\beta})H_N(a_{v\beta}^\dagger a_{j\alpha})|SS\rangle \quad (22)$$

$$= \delta_{ij}f_{av}^\beta + \langle a_\beta j\alpha || i\alpha v\beta \rangle. \quad (23)$$

$$A_{ai,vj} = \frac{1}{2}\left(1 + \frac{1}{S}\right)\delta_{ij}f_{av}^\beta + \frac{1}{2}\left(1 + \frac{1}{S}\right)(-K_{ai,vj}^{\alpha\alpha,\beta\beta} + K_{ai,vj}^{\beta\beta,\beta\beta}). \quad (24)$$

$$A_{ai,vj}^{1/2} = \frac{3}{2}\delta_{ij}f_{av}^\beta - \frac{1}{6}K_{ai,vj}^{\alpha\alpha,\beta\beta} + \frac{1}{6}K_{ai,vj}^{\beta\beta,\beta\beta} + \frac{4}{3}K_{ai,vj}^{\text{SF}}. \quad (25)$$

对于 CV-OO 块, 有

$$[A_+]_{ai,vw} = 0. \quad (26)$$

$$[A_0]_{ai,vw} = \frac{1}{2}\langle SS|(a_{i\alpha}^\dagger a_{a\alpha} - a_{i\beta}^\dagger a_{a\beta})H_N(a_{v\alpha}^\dagger a_{w\alpha})|SS\rangle \quad (27)$$

$$= \frac{1}{2}\delta_{vw}\langle SS|(a_{i\alpha}^\dagger a_{a\alpha} - a_{i\beta}^\dagger a_{a\beta})H_N|SS\rangle \quad (28)$$

$$= \frac{1}{2}\delta_{vw}(f_{ai}^\alpha - f_{ai}^\beta). \quad (29)$$

$$[A_-]_{ai,vw} = \langle SS|(a_{i\alpha}^\dagger a_{a\beta})H_N(a_{v\beta}^\dagger a_{w\alpha})|SS\rangle \quad (30)$$

$$= \langle a\beta w\alpha || i\alpha v\beta \rangle. \quad (31)$$

$$A_{ai,vw} = \frac{1}{2}\left(1 + \frac{1}{S}\right)\delta_{vw}(f_{ai}^\alpha - f_{ai}^\beta). \quad (32)$$

$[A_0]_{ai,vw}$ 的结果与 Li 和 Liu 文章中的 Table V 不同, 他们的结果是 0。

$$A_{ai,vw}^{1/2} = \frac{1}{6}\delta_{vw}(f_{ai}^\alpha - f_{ai}^\beta) + \frac{4}{3}K_{ai,vw}^{\text{SF}}. \quad (33)$$

这一项对于所有泛函均满足 $A^{\geq 1} = A^{1/2}$

对于 CV-OV 块, 有

$$[A_+]_{ai,bv} = 0. \quad (34)$$

$$[A_0]_{ai,bv} = \frac{1}{2} \langle SS | (a_{i\alpha}^\dagger a_{a\alpha} - a_{i\beta}^\dagger a_{a\beta}) H_N (a_{b\alpha}^\dagger a_{v\alpha}) | SS \rangle \quad (35)$$

$$= -\frac{1}{2} \delta_{ab} f_{vi}^\alpha + \frac{1}{2} [\langle a\alpha v\alpha || i\alpha b\alpha \rangle + \langle a\beta v\alpha || b\alpha i\beta \rangle] \quad (36)$$

$$= -\frac{1}{2} \delta_{ab} f_{vi}^\alpha + \frac{1}{2} [\langle a\alpha v\alpha || i\alpha b\alpha \rangle - \langle a\beta v\alpha || i\beta b\alpha \rangle]. \quad (37)$$

$$[A_-]_{ai,bv} = \langle SS | (a_{i\alpha}^\dagger a_{a\beta}) H_N (a_{b\beta}^\dagger a_{v\alpha}) | SS \rangle \quad (38)$$

$$= -\delta_{ab} f_{vi}^\alpha + \langle a\beta v\alpha || i\alpha b\beta \rangle. \quad (39)$$

$$A_{ai,bv} = -\frac{1}{2} (1 + \frac{1}{S}) \delta_{ab} f_{vi}^\alpha + \frac{1}{2} (1 + \frac{1}{S}) (K_{ai,bv}^{\alpha\alpha,\alpha\alpha} - K_{ai,bv}^{\beta\beta,\alpha\alpha}). \quad (40)$$

$$A_{ai,bv}^{1/2} = -\frac{3}{2} \delta_{ab} f_{vi}^\alpha + \frac{1}{6} K_{ai,bv}^{\alpha\alpha,\alpha\alpha} - \frac{1}{6} K_{ai,bv}^{\beta\beta,\alpha\alpha} + \frac{4}{3} K_{ai,bv}^{\text{SF}}. \quad (41)$$

对于 CO-CO 块, 有

$$[A_+]_{ui,vj} = 0. \quad (42)$$

$$[A_0]_{ui,vj} = \frac{1}{2} \langle SS | (-a_{i\beta}^\dagger a_{u\beta}) H_N (-a_{v\beta}^\dagger a_{j\beta}) | SS \rangle \quad (43)$$

$$= \frac{1}{2} \delta_{ij} f_{uv}^\beta - \frac{1}{2} \delta_{uv} f_{ji}^\beta + \frac{1}{2} \langle u\beta j\beta || i\beta v\beta \rangle. \quad (44)$$

$$[A_-]_{ui,vj} = \langle SS | (a_{i\alpha}^\dagger a_{u\beta}) H_N (a_{v\beta}^\dagger a_{j\alpha}) | SS \rangle \quad (45)$$

$$= \delta_{ij} f_{uv}^\beta - \delta_{uv} f_{ji}^\alpha + \langle u\beta j\alpha || i\alpha v\beta \rangle. \quad (46)$$

$$A_{ui,vj} = \frac{1}{2} (1 + \frac{1}{S}) \delta_{ij} f_{uv}^\beta - \frac{1}{2} (1 + \frac{1}{S}) \delta_{uv} f_{ji}^\beta + \frac{1}{2} (1 + \frac{1}{S}) K_{ui,vj}^{\beta\beta,\beta\beta}. \quad (47)$$

$$A_{ui,vj}^{1/2} = \frac{3}{2} \delta_{ij} f_{uv}^\beta - \frac{1}{6} \delta_{uv} f_{ji}^\beta - \frac{4}{3} \delta_{uv} f_{ji}^\alpha + \frac{1}{6} K_{ui,vj}^{\beta\beta,\beta\beta} + \frac{4}{3} K_{ui,vj}^{\text{SF}}. \quad (48)$$

对于 CO-OO 块, 有

$$[A_+]_{ui,vw} = 0. \quad (49)$$

$$[A_0]_{ui,vw} = \frac{1}{2} \langle SS | (-a_{i\beta}^\dagger a_{u\beta}) H_N (a_{v\alpha}^\dagger a_{w\alpha}) | SS \rangle \quad (50)$$

$$= \frac{1}{2} \delta_{vw} \langle SS | (-a_{i\beta}^\dagger a_{u\beta}) H_N | SS \rangle \quad (51)$$

$$= -\frac{1}{2} \delta_{vw} f_{ui}^\beta. \quad (52)$$

$$[A_-]_{ui,vw} = \langle SS | (a_{i\alpha}^\dagger a_{u\beta}) H_N (a_{v\beta}^\dagger a_{w\alpha}) | SS \rangle \quad (53)$$

$$= -\delta_{uv} f_{wi}^\alpha + \langle u\beta w\alpha || i\alpha v\beta \rangle. \quad (54)$$

$$A_{ui,vw} = -\frac{1}{2} (1 + \frac{1}{S}) \delta_{vw} f_{ui}^\beta. \quad (55)$$

在 Brillouin 条件下，这一项为零。

$$A_{ui,vw}^{1/2} = -\frac{1}{6}\delta_{vw}f_{ui}^\beta - \frac{4}{3}\delta_{uv}f_{wi}^\alpha + \frac{4}{3}K_{ui,vw}^{\text{SF}}. \quad (56)$$

注意这里实际上 uvw 三个轨道是同一个。

对于 CO-OV 块，有

$$[A_+]_{ui,bv} = 0. \quad (57)$$

$$[A_0]_{ui,bv} = \frac{1}{2} \langle SS | (-a_{i\beta}^\dagger a_{u\beta}) H_N (a_{b\alpha}^\dagger a_{v\alpha}) | SS \rangle \quad (58)$$

$$= \frac{1}{2} \langle u\beta v\alpha || b\alpha i\beta \rangle \quad (59)$$

$$= -\frac{1}{2} \langle u\beta v\alpha || i\beta b\alpha \rangle. \quad (60)$$

$$[A_-]_{ui,bv} = \langle SS | (a_{i\alpha}^\dagger a_{u\beta}) H_N (a_{b\beta}^\dagger a_{v\alpha}) | SS \rangle \quad (61)$$

$$= \langle u\beta v\alpha || i\alpha b\beta \rangle. \quad (62)$$

$$A_{ui,bv} = -\frac{1}{2} \left(1 + \frac{1}{S}\right) K_{ui,bv}^{\beta\beta,\alpha\alpha}. \quad (63)$$

$$A_{ui,bv}^{1/2} = -\frac{1}{6} K_{ui,bv}^{\beta\beta,\alpha\alpha} + \frac{4}{3} K_{ui,bv}^{\text{SF}}. \quad (64)$$

对于 OO-OV 块，有

$$[A_+]_{tu,vw} = 0. \quad (65)$$

$$[A_0]_{tu,vw} = \frac{1}{2} \langle SS | (a_{u\alpha}^\dagger a_{t\alpha}) H_N (a_{v\alpha}^\dagger a_{w\alpha}) | SS \rangle \quad (66)$$

$$= \frac{1}{2} \delta_{tu} \delta_{vw} \langle SS | H_N | SS \rangle \quad (67)$$

$$= 0. \quad (68)$$

$$[A_-]_{tu,vw} = \langle SS | (a_{u\alpha}^\dagger a_{t\beta}) H_N (a_{v\beta}^\dagger a_{w\alpha}) | SS \rangle \quad (69)$$

$$= \delta_{uw} f_{tv}^\beta - \delta_{tv} f_{wu}^\alpha + \langle t\beta w\alpha || u\alpha v\beta \rangle. \quad (70)$$

$$A_{tu,vw} = 0. \quad (71)$$

$$A_{tu,vw}^{1/2} = \frac{4}{3} \delta_{uw} f_{tv}^\beta - \frac{4}{3} \delta_{tv} f_{wu}^\alpha + \frac{4}{3} K_{tu,vw}^{\text{SF}}. \quad (72)$$

注意这四个轨道实际上是同一个，所以也是零。

对于 OO-OV 块，有

$$[A_+]_{tu,bv} = 0. \quad (73)$$

$$[A_0]_{tu,bv} = \frac{1}{2} \langle SS | (a_{u\alpha}^\dagger a_{t\alpha}) H_N (a_{b\alpha}^\dagger a_{v\alpha}) | SS \rangle \quad (74)$$

$$= \frac{1}{2} \delta_{tu} \langle SS | H_N (a_{b\alpha}^\dagger a_{v\alpha}) | SS \rangle \quad (75)$$

$$= \frac{1}{2} \delta_{tu} f_{vb}^\alpha. \quad (76)$$

$$[A_-]_{tu,bv} = \langle SS | (a_{u\alpha}^\dagger a_{t\beta}) H_N (a_{b\beta}^\dagger a_{v\alpha}) | SS \rangle \quad (77)$$

$$= \delta_{uv} f_{tb}^\beta + \langle t\beta v\alpha || u\alpha b\beta \rangle. \quad (78)$$

$$A_{tu,bv} = \frac{1}{2}(1 + \frac{1}{S})\delta_{tu}f_{vb}^\alpha. \quad (79)$$

在 Brillouin 条件下, 这一项为零。

$$A_{tu,bv}^{1/2} = \frac{1}{6}\delta_{tu}f_{vb}^\alpha + \frac{4}{3}\delta_{uv}f_{tb}^\beta + \frac{4}{3}K_{tu,bv}^{\text{SF}}. \quad (80)$$

对于 OV-OV 块, 有

$$[A_+]_{au,bv} = 0. \quad (81)$$

$$[A_0]_{au,bv} = \frac{1}{2} \langle SS | (a_{u\alpha}^\dagger a_{a\alpha}) H_N (a_{b\alpha}^\dagger a_{v\alpha}) | SS \rangle \quad (82)$$

$$= \frac{1}{2}\delta_{uv}f_{ab}^\alpha - \frac{1}{2}\delta_{ab}f_{vu}^\alpha + \frac{1}{2} \langle a\alpha v\alpha | | u\alpha b\alpha \rangle. \quad (83)$$

$$[A_-]_{au,bv} = \langle SS | (a_{u\alpha}^\dagger a_{a\beta}) H_N (a_{b\beta}^\dagger a_{v\alpha}) | SS \rangle \quad (84)$$

$$= \delta_{uv}f_{ab}^\beta - \delta_{ab}f_{vu}^\alpha + \langle a\alpha v\alpha | | u\beta b\alpha \rangle. \quad (85)$$

$$A_{au,bv} = \frac{1}{2}(1 + \frac{1}{S})\delta_{uv}f_{ab}^\alpha - \frac{1}{2}(1 + \frac{1}{S})\delta_{ab}f_{vu}^\alpha + \frac{1}{2}(1 + \frac{1}{S})K_{au,bv}^{\alpha\alpha,\alpha\alpha}. \quad (86)$$

$$A_{au,bv}^{1/2} = \frac{1}{6}\delta_{uv}f_{ab}^\alpha + \frac{4}{3}\delta_{uv}f_{ab}^\beta - \frac{3}{2}\delta_{ab}f_{vu}^\alpha + \frac{1}{6}K_{au,bv}^{\alpha\alpha,\alpha\alpha} + \frac{4}{3}K_{au,bv}^{\text{SF}}. \quad (87)$$

接下来看重叠矩阵 S 。对于 S_- , 有

$$[S_-]_{pq,rs} = \delta_{pr}\delta_{qs}. \quad (88)$$

接下来看重叠矩阵 S 。对于 CV-CV 块, 有

$$[S_+]_{ai,bj} = \delta_{ab}\delta_{ij}. \quad (89)$$

$$[S_0]_{ai,bj} = \delta_{ab}\delta_{ij}. \quad (90)$$

$$S_{ai,bj} = \delta_{ab}\delta_{ij}. \quad (91)$$

对于 CO-CO 块, 有

$$[S_+]_{ui,vj} = 0. \quad (92)$$

$$[S_0]_{ui,vj} = \frac{1}{2}\delta_{uv}\delta_{ij}. \quad (93)$$

$$S_{ui,vj} = \frac{1}{2}(1 + \frac{1}{S})\delta_{uv}\delta_{ij}. \quad (94)$$

对于 OO-OO 块, 有

$$[S_+]_{tu,vw} = 0. \quad (95)$$

$$[S_0]_{tu,vw} = \frac{1}{2}\delta_{tu}\delta_{vw}. \quad (96)$$

$$S_{tu,vw} = \frac{1}{2}(1 + \frac{1}{S})\delta_{tu}\delta_{vw}. \quad (97)$$

对于 OV-OV 块, 有

$$[S_+]_{au,bv} = 0. \quad (98)$$

$$[S_0]_{au,bv} = \frac{1}{2}\delta_{ab}\delta_{uv}. \quad (99)$$

$$S_{au,bv} = \frac{1}{2}(1 + \frac{1}{S})\delta_{ab}\delta_{uv}. \quad (100)$$

因此, 类似于 Chibueze 和 Visscher 的讨论, 同样将 OO-OO 块也分成 O1 和 O2 两部分, O2 部分的重叠矩阵就是零矩阵。O1 部分的重叠矩阵为

$$S^{O1} \propto J^{O1}, \quad (101)$$

其中 J 是全 1 矩阵。而 J 矩阵仅有一个非零特征值, 对应的特征向量为 $(1, 1, \dots, 1)^T$ 。因此, OO-OO 块中仅有一个线性无关的 CSF, 为 $|S, S-1\rangle$ 。后期需要对其进行投影。

1.2 单重态激发块的 A 和 S

注意实际上除了 CV 类型的激发之外, CO、OO、OV 类型的三重态和单重态激发其实是一致的 (最多只差相位), 因此我们只需要计算 CV 类型的单重态激发块的 A 和 S , 不妨记为 CV(0)。

A 矩阵

$$[A_{00}]_{ai,bj} = \frac{1}{2} \langle SS | (a_{i\alpha}^\dagger a_{a\alpha} + a_{i\beta}^\dagger a_{a\beta}) H_N (a_{b\alpha}^\dagger a_{j\alpha} + a_{b\beta}^\dagger a_{j\beta}) | SS \rangle \quad (102)$$

$$= \frac{1}{2} \delta_{ij} (f_{ab}^\alpha + f_{ab}^\beta) - \frac{1}{2} \delta_{ab} (f_{ji}^\alpha + f_{ji}^\beta) + \frac{1}{2} [\langle a\alpha j\alpha || i\alpha b\alpha \rangle + \langle a\alpha j\beta || i\alpha b\beta \rangle + \langle a\beta j\alpha || i\beta b\alpha \rangle + \langle a\beta j\beta || i\beta b\beta \rangle]. \quad (103)$$

$$A_{ai,bj} = \frac{1}{2} \delta_{ij} (f_{ab}^\alpha + f_{ab}^\beta) - \frac{1}{2} \delta_{ab} (f_{ji}^\alpha + f_{ji}^\beta) + K_{ai,bj}^S. \quad (104)$$

其中

$$K^S = \frac{1}{2} [K^{\alpha\alpha,\alpha\alpha} + K^{\alpha\alpha,\beta\beta} + K^{\beta\beta,\alpha\alpha} + K^{\beta\beta,\beta\beta}] = 2K^{00}. \quad (105)$$

重叠矩阵

$$[S_{00}]_{ai,bj} = \frac{1}{2} \langle SS | (a_{i\alpha}^\dagger a_{a\alpha} + a_{i\beta}^\dagger a_{a\beta}) (a_{b\alpha}^\dagger a_{j\alpha} + a_{b\beta}^\dagger a_{j\beta}) | SS \rangle \quad (106)$$

$$= \delta_{ab} \delta_{ij}. \quad (107)$$

$$S_{ai,bj} = \delta_{ab} \delta_{ij}. \quad (108)$$

1.3 单重态激发与三重态激发之间的耦合

CV(0)-CV 块

$$[A_{01}]_{ai,bj} = \frac{1}{2} \langle SS | (a_{i\alpha}^\dagger a_{a\alpha} + a_{i\beta}^\dagger a_{a\beta}) H_N (a_{b\alpha}^\dagger a_{j\alpha} - a_{b\beta}^\dagger a_{j\beta}) | SS \rangle \quad (109)$$

$$= \frac{1}{2} \delta_{ij} (f_{ab}^\alpha - f_{ab}^\beta) - \frac{1}{2} \delta_{ab} (f_{ji}^\alpha - f_{ji}^\beta) + \frac{1}{2} [\langle a\alpha j\alpha || i\alpha b\alpha \rangle - \langle a\alpha j\beta || i\alpha b\beta \rangle + \langle a\beta j\alpha || i\beta b\alpha \rangle - \langle a\beta j\beta || i\beta b\beta \rangle]. \quad (110)$$

$$A_{ai,bj} = -\frac{1}{2} \sqrt{\frac{S+1}{S}} \delta_{ij} (f_{ab}^\alpha - f_{ab}^\beta) + \frac{1}{2} \sqrt{\frac{S+1}{S}} \delta_{ab} (f_{ji}^\alpha - f_{ji}^\beta) - \frac{1}{2} \sqrt{\frac{S+1}{S}} [K_{ai,bj}^{\alpha\alpha,\alpha\alpha} - K_{ai,bj}^{\alpha\alpha,\beta\beta} + K_{ai,bj}^{\beta\beta,\alpha\alpha} - K_{ai,bj}^{\beta\beta,\beta\beta}]. \quad (111)$$

CV(0)-CO 块

$$[A_{01}]_{ai,vj} = \frac{1}{2} \langle SS | (a_{i\alpha}^\dagger a_{a\alpha} + a_{i\beta}^\dagger a_{a\beta}) H_N (-a_{v\beta}^\dagger a_{j\beta}) | SS \rangle \quad (112)$$

$$= -\frac{1}{2} \delta_{ij} f_{av}^\beta + \frac{1}{2} [-\langle a\alpha j\beta || v\beta i\alpha \rangle - \langle a\beta j\beta || i\beta v\beta \rangle]. \quad (113)$$

$$A_{ai,vj} = \frac{1}{2}\sqrt{\frac{S+1}{S}}\delta_{ij}f_{av}^\beta + \frac{1}{2}\sqrt{\frac{S+1}{S}}[K_{ai,vj}^{\alpha\alpha,\beta\beta} + K_{ai,vj}^{\beta\beta,\beta\beta}]. \quad (114)$$

CV(0)-OO 块

$$[A_{01}]_{ai,vw} = \frac{1}{2}\langle SS|(a_{i\alpha}^\dagger a_{a\alpha} + a_{i\beta}^\dagger a_{a\beta})H_N(a_{v\alpha}^\dagger a_{w\alpha})|SS\rangle \quad (115)$$

$$= \frac{1}{2}\delta_{vw}\langle SS|(a_{i\alpha}^\dagger a_{a\alpha} + a_{i\beta}^\dagger a_{a\beta})H_N|SS\rangle \quad (116)$$

$$= \frac{1}{2}\delta_{vw}(f_{ai}^\alpha + f_{ai}^\beta). \quad (117)$$

$$A_{ai,vw} = -\frac{1}{2}\sqrt{\frac{S+1}{S}}\delta_{vw}(f_{ai}^\alpha + f_{ai}^\beta). \quad (118)$$

CV(0)-OV 块

$$[A_{01}]_{ai,bv} = \frac{1}{2}\langle SS|(a_{i\alpha}^\dagger a_{a\alpha} + a_{i\beta}^\dagger a_{a\beta})H_N(a_{b\alpha}^\dagger a_{v\alpha})|SS\rangle \quad (119)$$

$$= -\frac{1}{2}\delta_{ab}f_{vi}^\alpha + \frac{1}{2}[\langle a\alpha v\alpha||i\alpha b\alpha\rangle + \langle a\beta v\alpha||b\alpha i\beta\rangle]. \quad (120)$$

$$A_{ai,bv} = \frac{1}{2}\sqrt{\frac{S+1}{S}}\delta_{ab}f_{vi}^\alpha - \frac{1}{2}\sqrt{\frac{S+1}{S}}[K_{ai,bv}^{\alpha\alpha,\alpha\alpha} + K_{ai,bv}^{\beta\beta,\alpha\alpha}]. \quad (121)$$

1.4 总的结果

考虑将归一化合并入 A 矩阵中, 即

$$A_{pq,rs} = \frac{A_{pq,rs}}{\sqrt{S_{pq,pq}S_{rs,rs}}}. \quad (122)$$

表 1: $S_f = S_i$ 中使用的 A 矩阵

Block	Matrix elements
CV-CV	$A_{ai,bj} = \frac{1}{2}\delta_{ij}[(1 - \frac{1}{S})f_{ab}^\alpha + (1 + \frac{1}{S})f_{ab}^\beta] - \frac{1}{2}\delta_{ab}[(1 + \frac{1}{S})f_{ji}^\alpha + (1 - \frac{1}{S})f_{ji}^\beta] + (1 + \frac{1}{S})K_{ai,bj}^T - \frac{1}{S}K_{ai,bj}^{\text{SF}}$
CV-CO	$A_{ai,vj} = \sqrt{\frac{S+1}{2S}}\delta_{ij}f_{av}^\beta + \sqrt{\frac{S+1}{2S}}(-K_{ai,vj}^{\alpha\alpha,\beta\beta} + K_{ai,vj}^{\beta\beta,\beta\beta})$
CV-OO	$A_{ai,vw} = \sqrt{\frac{S+1}{2S}}\delta_{vw}(\textcolor{red}{f}_{ai}^\alpha - \textcolor{red}{f}_{ai}^\beta)$
CV-OV	$A_{ai,bv} = -\sqrt{\frac{S+1}{2S}}\delta_{ab}f_{vi}^\alpha + \sqrt{\frac{S+1}{2S}}(K_{ai,bv}^{\alpha\alpha,\alpha\alpha} - K_{ai,bv}^{\beta\beta,\alpha\alpha})$
CO-CO	$A_{ui,vj} = \delta_{ij}f_{uv}^\beta - \delta_{uv}f_{ji}^\beta + K_{ui,vj}^{\beta\beta,\beta\beta}$
CO-OO	$A_{ui,vw} = -\delta_{vw}f_{ui}^\beta$
CO-OV	$A_{ui,bv} = -K_{ui,bv}^{\beta\beta,\alpha\alpha}$
OO-OO	$A_{tu,vw} = 0$
OO-OV	$A_{tu,bv} = \delta_{tu}f_{vb}^\alpha$
OV-OV	$A_{au,bv} = \delta_{uv}f_{ab}^\alpha - \delta_{ab}f_{vu}^\alpha + K_{au,bv}^{\alpha\alpha,\alpha\alpha}$
CV(0)-CV(0)	$A_{ai,bj} = \frac{1}{2}\delta_{ij}(f_{ab}^\alpha + f_{ab}^\beta) - \frac{1}{2}\delta_{ab}(f_{ji}^\alpha + f_{ji}^\beta) + K_{ai,bj}^S$
CV(0)-CV	$A_{ai,bj} = -\frac{1}{2}\sqrt{\frac{S+1}{S}}\delta_{ij}(f_{ab}^\alpha - f_{ab}^\beta) + \frac{1}{2}\sqrt{\frac{S+1}{S}}\delta_{ab}(f_{ji}^\alpha - f_{ji}^\beta) - \frac{1}{2}\sqrt{\frac{S+1}{S}}[K_{ai,bj}^{\alpha\alpha,\alpha\alpha} - K_{ai,bj}^{\alpha\alpha,\beta\beta} + K_{ai,bj}^{\beta\beta,\alpha\alpha} - K_{ai,bj}^{\beta\beta,\beta\beta}]$
CV(0)-CO	$A_{ai,vj} = \frac{1}{\sqrt{2}}\delta_{ij}f_{av}^\beta + \frac{1}{\sqrt{2}}(K_{ai,vj}^{\alpha\alpha,\beta\beta} + K_{ai,vj}^{\beta\beta,\beta\beta})$
CV(0)-OO	$A_{ai,vw} = -\frac{1}{\sqrt{2}}\delta_{vw}(f_{ai}^\alpha + f_{ai}^\beta)$
CV(0)-OV	$A_{ai,bv} = \frac{1}{\sqrt{2}}\delta_{ab}f_{vi}^\alpha - \frac{1}{\sqrt{2}}(K_{ai,bv}^{\alpha\alpha,\alpha\alpha} + K_{ai,bv}^{\beta\beta,\alpha\alpha})$

定义

$$f^0 = \frac{1}{2}(f^\alpha + f^\beta), \quad (123)$$

$$f^z = \frac{1}{2}(f^\alpha - f^\beta), \quad (124)$$

$$K^S = \frac{1}{2}[K^{\alpha\alpha,\alpha\alpha} + K^{\alpha\alpha,\beta\beta} + K^{\beta\beta,\alpha\alpha} + K^{\beta\beta,\beta\beta}] = 2J - K, \quad (125)$$

$$K^T = \frac{1}{2}[K^{\alpha\alpha,\alpha\alpha} - K^{\alpha\alpha,\beta\beta} - K^{\beta\beta,\alpha\alpha} + K^{\beta\beta,\beta\beta}] = -K, \quad (126)$$

$$K^P = \frac{1}{2}[K^{\alpha\alpha,\alpha\alpha} + K^{\alpha\alpha,\beta\beta} - K^{\beta\beta,\alpha\alpha} - K^{\beta\beta,\beta\beta}]. \quad (127)$$

程序实现版本 (Method 0, 当不包含 OO 时则退化到 S-TDA):

表 2: $S_f = S_i \geq 1$ 的 S-TDA 使用的 A 矩阵

Block	Matrix elements
CV-CV	$A_{ia,jb} = \delta_{ij}(f_{ab}^0 - \frac{1}{S}f_{ab}^z) - \delta_{ab}(f_{ji}^0 + \frac{1}{S}f_{ji}^z) + (1 + \frac{1}{S})K_{ai,bj}^T - \frac{1}{S}K_{ai,bj}^{\text{SF}}$
CV-CO	$A_{ia,jv} = \sqrt{\frac{S+1}{2S}}\delta_{ij}f_{av}^\beta + \sqrt{\frac{S+1}{2S}}(-K_{ai,vj}^{\alpha\alpha,\beta\beta} + K_{ai,vj}^{\beta\beta,\beta\beta})$
CV-OO	$A_{ia,wv} = 2\sqrt{\frac{S+1}{2S}}\delta_{vw}f_{ai}^z$
CV-OV	$A_{ia,vb} = -\sqrt{\frac{S+1}{2S}}\delta_{ab}f_{vi}^\alpha + \sqrt{\frac{S+1}{2S}}(K_{ai,bv}^{\alpha\alpha,\alpha\alpha} - K_{ai,bv}^{\beta\beta,\alpha\alpha})$
CO-CO	$A_{iu,jv} = \delta_{ij}f_{uv}^\beta - \delta_{uv}f_{ji}^\beta + K_{ui,vj}^{\beta\beta,\beta\beta}$
CO-OO	$A_{iu,wv} = -\delta_{vw}f_{ui}^\beta$
CO-OV	$A_{iu,vb} = -K_{ui,bv}^{\beta\beta,\alpha\alpha}$
OO-OO	$A_{ut,wv} = 0$
OO-OV	$A_{ut,vb} = \delta_{tu}f_{vb}^\alpha$
OV-OV	$A_{ua,vb} = \delta_{uv}f_{ab}^\alpha - \delta_{ab}f_{vu}^\alpha + K_{au,bv}^{\alpha\alpha,\alpha\alpha}$
CV(0)-CV(0)	$A_{ia,jb} = \delta_{ij}f_{ab}^0 - \delta_{ab}f_{ji}^0 + K_{ai,bj}^S$
CV(0)-CV	$A_{ia,jb} = -\sqrt{\frac{S+1}{S}}\delta_{ij}f_{ab}^z + \sqrt{\frac{S+1}{S}}\delta_{ab}f_{ji}^z - \sqrt{\frac{S+1}{S}}K_{ai,bj}^P$
CV(0)-CO	$A_{ia,jv} = \frac{1}{\sqrt{2}}\delta_{ij}f_{av}^\beta + \frac{1}{\sqrt{2}}(K_{ai,vj}^{\alpha\alpha,\beta\beta} + K_{ai,vj}^{\beta\beta,\beta\beta})$
CV(0)-OO	$A_{ia,wv} = -\sqrt{2}\delta_{vw}f_{ai}^0$
CV(0)-OV	$A_{ia,vb} = \frac{1}{\sqrt{2}}\delta_{ab}f_{vi}^\alpha - \frac{1}{\sqrt{2}}(K_{ai,bv}^{\alpha\alpha,\alpha\alpha} + K_{ai,bv}^{\beta\beta,\alpha\alpha})$

1.5 OO 相关矩阵元的归一化

从广义本征方程角度来看一下 OO 相关矩阵元的归一化问题。由于 O2 部分的 A 矩阵和 S 矩阵都是零，所以不考虑它们的贡献。这里 OO 就指代所有的 O1 (共 $2S$ 个)。将所有激发分为 OO 型和其他型 (rest)。则 A 矩阵和 S 矩阵形如

$$A = \begin{pmatrix} 0 & A_{\text{OO}-\text{rest}} \\ A_{\text{rest}-\text{OO}} & A_{\text{rest}-\text{rest}} \end{pmatrix}, \quad S = \begin{pmatrix} \frac{S+1}{2S}J & 0 \\ 0 & S_{\text{rest}-\text{rest}} \end{pmatrix}. \quad (128)$$

其中 $A_{\text{OO}-\text{rest}}$ 和 $A_{\text{rest}-\text{OO}}$ 与 OO 无关，此外 $S_{\text{rest}-\text{rest}}$ 是对角的。

将 S 矩阵进行特征分解，得到 $S = UsU^\dagger$ ，其中 s 是对角矩阵。由于 J 矩阵仅有一个非零特征值 $2S$ ，因此所有非零的 s 为

$$s = \text{diag}(S + 1, s_{\text{rest}}, \dots, s_{\text{rest}}). \quad (129)$$

对应的 U 矩阵为

$$U = \begin{pmatrix} \frac{1}{\sqrt{2S}} & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{\sqrt{2S}} & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}. \quad (130)$$

于是有

$$X = U s^{-1/2} = \begin{pmatrix} \frac{1}{\sqrt{2S(S+1)}} & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{\sqrt{2S(S+1)}} & 0 & \cdots & 0 \\ 0 & s_{\text{rest}}^{-1/2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & s_{\text{rest}}^{-1/2} \end{pmatrix}. \quad (131)$$

因此变换后的 A 矩阵为

$$A' = X^\dagger A X = \begin{pmatrix} 0 & \frac{1}{\sqrt{2S(S+1)}} \mathbf{1}^T A_{\text{OO}-\text{rest}} s_{\text{rest}}^{-1/2} \\ \frac{1}{\sqrt{2S(S+1)}} s_{\text{rest}}^{-1/2} A_{\text{rest}-\text{OO}} \mathbf{1} & s_{\text{rest}}^{-1/2} A_{\text{rest}-\text{rest}} s_{\text{rest}}^{-1/2} \end{pmatrix}. \quad (132)$$

考虑到 $A_{\text{OO}-\text{rest}}$ 和 $A_{\text{rest}-\text{OO}}$ 的矩阵元与具体的 OO 指标无关（即各行/各列元素相同），求和会产生一个 $2S$ 的因子。定义 $\bar{A}_{\text{OO}-\text{rest}}$ 和 $\bar{A}_{\text{rest}-\text{OO}}$ 为坍塌后的单行/单列矩阵，则可以进一步化简为：

$$A' = X^\dagger A X = \begin{pmatrix} 0 & \sqrt{\frac{2S}{S+1}} \bar{A}_{\text{OO}-\text{rest}} s_{\text{rest}}^{-1/2} \\ \sqrt{\frac{2S}{S+1}} s_{\text{rest}}^{-1/2} \bar{A}_{\text{rest}-\text{OO}} & s_{\text{rest}}^{-1/2} A_{\text{rest}-\text{rest}} s_{\text{rest}}^{-1/2} \end{pmatrix}. \quad (133)$$

因此表中 OO 块相关的结果是正确的。

2 $S_f = S_i = \frac{1}{2}$ 的情况

先考虑把 S-TDA 中的矩阵 Table 2 强行做 $S = 1/2$ 的限制，即

表 3: 将 $S \geq 1$ 的结果延拓到 $S = 1/2$ 得到的 A 矩阵

Block	Matrix elements
CV-CV	$A_{ia,jb} = \delta_{ij}(f_{ab}^0 - 2f_{ab}^z) - \delta_{ab}(f_{ji}^0 + 2f_{ji}^z) + 3K_{ai,bj}^T - 2K_{ai,bj}^{\text{SF}}$
CV-CO	$A_{ia,jv} = \sqrt{\frac{3}{2}}\delta_{ij}f_{av}^\beta + \sqrt{\frac{3}{2}}(-K_{ai,vj}^{\alpha\alpha,\beta\beta} + K_{ai,vj}^{\beta\beta,\beta\beta})$
CV-OO	$A_{ia,wv} = \sqrt{6}\delta_{vw}f_{ai}^z$
CV-OV	$A_{ia,vb} = -\sqrt{\frac{3}{2}}\delta_{ab}f_{vi}^\alpha + \sqrt{\frac{3}{2}}(K_{ai,bv}^{\alpha\alpha,\alpha\alpha} - K_{ai,bv}^{\beta\beta,\alpha\alpha})$
CO-CO	$A_{iu,jv} = \delta_{ij}f_{uv}^\beta - \delta_{uv}f_{ji}^\beta + K_{ui,vj}^{\beta\beta,\beta\beta}$
CO-OO	$A_{iu,wv} = -\delta_{vw}f_{ui}^\beta$
CO-OV	$A_{iu,vb} = -K_{ui,bv}^{\beta\beta,\alpha\alpha}$
OO-OO	$A_{ut,wv} = 0$
OO-OV	$A_{ut,vb} = \delta_{tu}f_{vb}^\alpha$
OV-OV	$A_{ua,vb} = \delta_{uv}f_{ab}^\alpha - \delta_{ab}f_{vu}^\alpha + K_{au,bv}^{\alpha\alpha,\alpha\alpha}$
CV(0)-CV(0)	$A_{ia,jb} = \delta_{ij}f_{ab}^0 - \delta_{ab}f_{ji}^0 + K_{ai,bj}^S$
CV(0)-CV	$A_{ia,jb} = -\sqrt{3}\delta_{ij}f_{ab}^z + \sqrt{3}\delta_{ab}f_{ji}^z - \sqrt{3}K_{ai,bj}^P$
CV(0)-CO	$A_{ia,jv} = \frac{1}{\sqrt{2}}\delta_{ij}f_{av}^\beta + \frac{1}{\sqrt{2}}(K_{ai,vj}^{\alpha\alpha,\beta\beta} + K_{ai,vj}^{\beta\beta,\beta\beta})$
CV(0)-OO	$A_{ia,wv} = -\sqrt{2}\delta_{vw}f_{ai}^0$
CV(0)-OV	$A_{ia,vb} = \frac{1}{\sqrt{2}}\delta_{ab}f_{vi}^\alpha - \frac{1}{\sqrt{2}}(K_{ai,bv}^{\alpha\alpha,\alpha\alpha} + K_{ai,bv}^{\beta\beta,\alpha\alpha})$

再按照 Li 和 Liu 文章中的 (103), M 矩阵的形式为

$$\begin{pmatrix} M_{00} & -\sqrt{3}M_{01} \\ -\sqrt{3}M_{10} & -\frac{2}{3}M_+ + \frac{1}{3}M_0 + \frac{4}{3}M_- \end{pmatrix}. \quad (134)$$

计算得到是

表 4: 根据 $S = 1/2$ 原始公式得到的 S-TDA 的 A 矩阵

Block	Matrix elements
CV-CV	$A_{ia,jb} = \delta_{ij}(f_{ab}^0 - 2f_{ab}^z) - \delta_{ab}(f_{ji}^0 + 2f_{ji}^z) + \frac{1}{3}K_{ai,bj}^T + \frac{2}{3}K_{ai,bj}^{\text{SF}}$
CV-CO	$A_{ia,jv} = \sqrt{\frac{3}{2}}\delta_{ij}f_{av}^\beta - \frac{1}{6}\sqrt{\frac{2}{3}}K_{ai,vb}^{\alpha\alpha,\beta\beta} + \frac{1}{6}\sqrt{\frac{2}{3}}K_{ai,vj}^{\beta\beta,\beta\beta} + \frac{4}{3}\sqrt{\frac{2}{3}}K_{ai,vj}^{\text{SF}}$
CV-OO	$A_{ia,wv} = \frac{1}{3}\sqrt{\frac{2}{3}}\delta_{vw}f_{ai}^z + \frac{4}{3}\sqrt{\frac{2}{3}}K_{ai,wv}^{\text{SF}}$
CV-OV	$A_{ia,vb} = -\sqrt{\frac{3}{2}}\delta_{ab}f_{vi}^\alpha + \frac{1}{6}\sqrt{\frac{2}{3}}K_{ai,bv}^{\alpha\alpha,\alpha\alpha} - \frac{1}{6}\sqrt{\frac{2}{3}}K_{ai,bv}^{\beta\beta,\alpha\alpha} + \frac{4}{3}\sqrt{\frac{2}{3}}K_{ai,bv}^{\text{SF}}$
CO-CO	$A_{iu,jv} = \delta_{ij}f_{uv}^\beta - \frac{1}{9}\delta_{uv}f_{ji}^\beta - \frac{8}{9}\delta_{uv}f_{ji}^\alpha + \frac{1}{9}K_{ui,vj}^{\beta\beta,\beta\beta} + \frac{8}{9}K_{ui,vj}^{\text{SF}}$
CO-OO	$A_{iu,wv} = -\frac{1}{9}\delta_{vw}f_{ui}^\beta - \frac{8}{9}\delta_{uv}f_{wi}^\alpha + \frac{8}{9}K_{ui,wv}^{\text{SF}}$
CO-OV	$A_{iu,vb} = -\frac{1}{9}K_{ui,bv}^{\beta\beta,\alpha\alpha} + \frac{8}{9}K_{ui,bv}^{\text{SF}}$
OO-OO	$A_{ut,wv} = \frac{8}{9}\delta_{uw}f_{tv}^\beta - \frac{8}{9}\delta_{tv}f_{wu}^\alpha + \frac{8}{9}K_{tu,vw}^{\text{SF}}$
OO-OV	$A_{ut,vb} = \frac{1}{9}\delta_{tu}f_{vb}^\alpha + \frac{8}{9}\delta_{uv}f_{tb}^\beta + \frac{8}{9}K_{tu,bv}^{\text{SF}}$
OV-OV	$A_{ua,vb} = \frac{1}{9}\delta_{uv}f_{ab}^\alpha + \frac{8}{9}\delta_{uv}f_{ab}^\beta - \delta_{ab}f_{vu}^\alpha + \frac{1}{9}K_{au,bv}^{\alpha\alpha,\alpha\alpha} + \frac{8}{9}K_{au,bv}^{\text{SF}}$
CV(0)-CV(0)	$A_{ia,jb} = \delta_{ij}f_{ab}^0 - \delta_{ab}f_{ji}^0 + K_{ai,bj}^S$
CV(0)-CV	$A_{ia,jb} = -\sqrt{3}\delta_{ij}f_{ab}^z + \sqrt{3}\delta_{ab}f_{ji}^z - \sqrt{3}K_{ai,bj}^P$
CV(0)-CO	$A_{ia,jv} = \frac{1}{\sqrt{2}}\delta_{ij}f_{av}^\beta + \frac{1}{\sqrt{2}}(K_{ai,vj}^{\alpha\alpha,\beta\beta} + K_{ai,vj}^{\beta\beta,\beta\beta})$
CV(0)-OO	$A_{ia,wv} = -\sqrt{2}\delta_{vw}f_{ai}^0$
CV(0)-OV	$A_{ia,vb} = \frac{1}{\sqrt{2}}\delta_{ab}f_{vi}^\alpha - \frac{1}{\sqrt{2}}(K_{ai,bv}^{\alpha\alpha,\alpha\alpha} + K_{ai,bv}^{\beta\beta,\alpha\alpha})$

可以看到, 对于 HF, Table 3 和 Table 4 是一样的, 但是对于泛函, 它们一般是不同的。

为了使它们两者相同, 需要满足一些要求 (**TODO: 检查一下充分必要性**)。比如对于 CV-CV 块, 需要满足

$$K_{ai,bj}^T = K_{ai,bj}^{\text{SF}}. \quad (135)$$

对于 CV-OO、CO-OO、OO-OO 和 OO-OV 块, 需要满足

$$f_{pq}^\alpha - f_{pq}^\beta = K_{pq,tt}^{\text{SF}}. \quad (136)$$

对于 CO-OV 块, 需要满足

$$K_{ti,bt}^{\beta\beta,\alpha\alpha} = -K_{ti,bt}^{\text{SF}}. \quad (137)$$

对于 CO-CO 块, 结合零激发能定理, 需要满足

$$K_{ti,tj}^{\beta\beta,\beta\beta} = K_{ti,tj}^{\text{SF}} - K_{ji,tt}^{\text{SF}}. \quad (138)$$

对于 OV-OV 块, 结合零激发能定理, 需要满足

$$K_{at,bt}^{\alpha\alpha,\alpha\alpha} = K_{at,bt}^{\text{SF}} - K_{ab,tt}^{\text{SF}}. \quad (139)$$

因此, 不失一般地, 将 K^{SF} 变成某一个一般矩阵元, 记为 K^{Ref} 。并且假设

$$f_{pq}^\alpha - f_{pq}^\beta = 2f_{pq}^z = \sum_t K_{pq,tt}^{\text{Ref}}, \quad (140)$$

以及

$$K_{pq,rs}^{\alpha\alpha,\alpha\alpha} = K_{pq,rs}^{\beta\beta,\beta\beta} = K_{pq,rs}^{\text{Ref}} - K_{pr,qs}^{\text{Ref}}, \quad (141)$$

和

$$K_{pq,rs}^{\alpha\alpha,\beta\beta} = K_{pq,rs}^{\beta\beta,\alpha\alpha} = -K_{pr,qs}^{\text{Ref}}. \quad (142)$$

因此，将所有结果用 f^0 和 K^{ref} 表示为

表 5: $S_f = S_i$ 使用的 A 矩阵

Block	Matrix elements
CV-CV	$A_{ia,jb} = \delta_{ij}(f_{ab}^0 - \frac{1}{S}f_{ab}^z) - \delta_{ab}(f_{ji}^0 + \frac{1}{S}f_{ji}^z) + K_{ai,bj}^{\text{Ref}}$
CV-CO	$A_{ia,jv} = \sqrt{\frac{S+1}{2S}}\delta_{ij}(f_{av}^0 - f_{av}^z) + \sqrt{\frac{S+1}{2S}}K_{ai,vj}^{\text{Ref}}$
CV-OO	$A_{ia,wv} = 2\sqrt{\frac{S+1}{2S}}\delta_{vw}f_{ai}^z$
CV-OV	$A_{ia,vb} = -\sqrt{\frac{S+1}{2S}}\delta_{ab}(f_{vi}^0 + f_{vi}^z) + \sqrt{\frac{S+1}{2S}}K_{ai,bv}^{\text{Ref}}$
CO-CO	$A_{iu,jv} = \delta_{ij}(f_{uv}^0 - f_{uv}^z) - \delta_{uv}(f_{ji}^0 - f_{ji}^z) + K_{ui,vj}^{\text{Ref}} - K_{uv,ij}^{\text{Ref}}$
CO-OO	$A_{iu,wv} = -\delta_{vw}(f_{ui}^0 - f_{ui}^z)$
CO-OV	$A_{iu,vb} = K_{ub,iv}^{\text{Ref}}$
OO-OO	$A_{ut,wv} = 0$
OO-OV	$A_{ut,vb} = \delta_{tu}(f_{vb}^0 + f_{vb}^z)$
OV-OV	$A_{ua,vb} = \delta_{uv}(f_{ab}^0 + f_{ab}^z) - \delta_{ab}(f_{vu}^0 + f_{vu}^z) + K_{au,bv}^{\text{Ref}} - K_{ab,uv}^{\text{Ref}}$
CV(0)-CV(0)	$A_{ia,jb} = \delta_{ij}f_{ab}^0 - \delta_{ab}f_{ji}^0 + K_{ai,bj}^S$
CV(0)-CV	$A_{ia,jb} = -\sqrt{\frac{S+1}{S}}\delta_{ij}f_{ab}^z + \sqrt{\frac{S+1}{S}}\delta_{ab}f_{ji}^z - \sqrt{\frac{S+1}{S}}K_{ai,bj}^P$
CV(0)-CO	$A_{ia,jv} = \frac{1}{\sqrt{2}}\delta_{ij}f_{av}^\beta + \frac{1}{\sqrt{2}}(K_{ai,vj}^{\alpha\alpha,\beta\beta} + K_{ai,vj}^{\beta\beta,\beta\beta})$
CV(0)-OO	$A_{ia,wv} = -\sqrt{2}\delta_{vw}f_{ai}^0$
CV(0)-OV	$A_{ia,vb} = \frac{1}{\sqrt{2}}\delta_{ab}f_{vi}^\alpha - \frac{1}{\sqrt{2}}(K_{ai,bv}^{\alpha\alpha,\alpha\alpha} + K_{ai,bv}^{\beta\beta,\alpha\alpha})$

在这一前提下，提出 Method 4: 令

$$f_{pq}^z = \frac{1}{2} \sum_t K_{pq,tt}^{\text{Ref}}, \quad (143)$$

还有

$$f_{pq}^0 = f_{pq}^\alpha - f_{pq}^z, \quad (144)$$

以及

$$K_{pq,rs}^{\text{Ref}} = K_{pq,rs}^T (D^z = 0). \quad (145)$$

即所有的矩阵元（除了 CV(0) 相关），均由 f^α 和 K^T 表示。

Block	Matrix elements
CO-CO	$A_{iu,jv} = \delta_{ij}(f_{uv}^0 - f_{uv}^z) - \delta_{uv}(f_{ji}^0 - f_{ji}^z) + K_{ui,vj}^{\text{Ref}} - K_{uv,ij}^{\text{Ref}}$
CO-CV	$A_{iu,jb} = \sqrt{\frac{S+1}{2S}}\delta_{ij}(f_{ub}^0 - f_{ub}^z) + \sqrt{\frac{S+1}{2S}}K_{ui,bj}^{\text{Ref}}$
CO-OO	$A_{iu,wv} = -\delta_{vw}(f_{ui}^0 - f_{ui}^z)$
CO-OV	$A_{iu,vb} = K_{ub,iv}^{\text{Ref}}$
CO-CV(0)	$A_{iu,jb} = \frac{1}{\sqrt{2}}\delta_{ij}f_{ub}^\beta + \frac{1}{\sqrt{2}}(K_{ui,bj}^{\beta\beta,\alpha\alpha} + K_{ui,bj}^{\beta\beta,\beta\beta})$
CV-CO	$A_{ia,jv} = \sqrt{\frac{S+1}{2S}}\delta_{ij}(f_{av}^0 - f_{av}^z) + \sqrt{\frac{S+1}{2S}}K_{ai,vj}^{\text{Ref}}$
CV-CV	$A_{ia,jb} = \delta_{ij}(f_{ab}^0 - \frac{1}{S}f_{ab}^z) - \delta_{ab}(f_{ji}^0 + \frac{1}{S}f_{ji}^z) + K_{ai,bj}^{\text{Ref}}$
CV-OO	$A_{ia,wv} = 2\sqrt{\frac{S+1}{2S}}\delta_{vw}f_{ai}^z$
CV-OV	$A_{ia,vb} = -\sqrt{\frac{S+1}{2S}}\delta_{ab}(f_{vi}^0 + f_{vi}^z) + \sqrt{\frac{S+1}{2S}}K_{ai,bv}^{\text{Ref}}$
CV-CV(0)	$A_{ia,jb} = -\sqrt{\frac{S+1}{S}}\delta_{ij}f_{ab}^z + \sqrt{\frac{S+1}{S}}\delta_{ab}f_{ji}^z - \sqrt{\frac{S+1}{S}}K_{ai,bj}^{?P?}$
OO-CO	$A_{ut,jv} = -\delta_{ut}(f_{jv}^0 - f_{jv}^z)$
OO-CV	$A_{ut,jb} = 2\sqrt{\frac{S+1}{2S}}\delta_{ut}f_{jb}^z$
OO-OO	$A_{ut,wv} = 0$
OO-OV	$A_{ut,vb} = \delta_{tu}(f_{vb}^0 + f_{vb}^z)$
OO-CV(0)	$A_{ut,jb} = -\sqrt{2}\delta_{ut}f_{jb}^0$
OV-CO	$A_{ua,jv} = K_{av,uj}^{\text{Ref}}$
OV-CV	$A_{ua,jb} = -\sqrt{\frac{S+1}{2S}}\delta_{ab}(f_{ju}^0 + f_{ju}^z) + \sqrt{\frac{S+1}{2S}}K_{au,bj}^{\text{Ref}}$
OV-OO	$A_{ua,wv} = \delta_{vw}(f_{au}^0 + f_{au}^z)$
OV-OV	$A_{ua,vb} = \delta_{uv}(f_{ab}^0 + f_{ab}^z) - \delta_{ab}(f_{vu}^0 + f_{vu}^z) + K_{au,bv}^{\text{Ref}} - K_{ab,uv}^{\text{Ref}}$
OV-CV(0)	$A_{ua,jb} = \frac{1}{\sqrt{2}}\delta_{ab}f_{ju}^\alpha - \frac{1}{\sqrt{2}}(K_{au,bj}^{\alpha\alpha,\alpha\alpha} + K_{au,bj}^{\alpha\alpha,\beta\beta})$
CV(0)-CO	$A_{ia,jv} = \frac{1}{\sqrt{2}}\delta_{ij}f_{av}^\beta + \frac{1}{\sqrt{2}}(K_{ai,vj}^{\alpha\alpha,\beta\beta} + K_{ai,vj}^{\beta\beta,\beta\beta})$
CV(0)-CV	$A_{ia,jb} = -\sqrt{\frac{S+1}{S}}\delta_{ij}f_{ab}^z + \sqrt{\frac{S+1}{S}}\delta_{ab}f_{ji}^z - \sqrt{\frac{S+1}{S}}K_{ai,bj}^P$
CV(0)-OO	$A_{ia,wv} = -\sqrt{2}\delta_{vw}f_{ai}^0$
CV(0)-OV	$A_{ia,vb} = \frac{1}{\sqrt{2}}\delta_{ab}f_{vi}^\alpha - \frac{1}{\sqrt{2}}(K_{ai,bv}^{\alpha\alpha,\alpha\alpha} + K_{ai,bv}^{\beta\beta,\alpha\alpha})$
CV(0)-CV(0)	$A_{ia,jb} = \delta_{ij}f_{ab}^0 - \delta_{ab}f_{ji}^0 + K_{ai,bj}^S$

3 $S_f = S_i - 1$ 的情况

Block	Matrix elements
CO-CO	$A_{iu,jv} = \delta_{ij}(f_{uv}^0 - f_{uv}^z) - \delta_{uv}(f_{ji}^0 + f_{ji}^z) - \frac{2}{2S-1}\delta_{uv}f_{ji}^z + K_{ui,vj}^{\text{Ref}} + \frac{1}{2S-1}K_{uv,ij}^{\text{Ref}}$
CO-CV	$A_{iu,jb} = \sqrt{\frac{2S+1}{2S}}\delta_{ij}(f_{ub}^0 - f_{ub}^z) + \sqrt{\frac{2S+1}{2S}}K_{ui,bj}^{\text{Ref}}$
CO-OO	$A_{iu,wv} = -\sqrt{\frac{2S}{2S-1}}\delta_{uv}(f_{wi}^0 + f_{wi}^z) + \frac{1}{\sqrt{2S(2S-1)}}\delta_{vw}(f_{ui}^0 - f_{ui}^z) + \sqrt{\frac{2S}{2S-1}}K_{ui,vw}^{\text{Ref}}$
CO-OV	$A_{iu,vb} = \frac{2S}{2S-1}K_{ui,bv}^{\text{Ref}} - \frac{1}{2S-1}K_{ub,iv}^{\text{Ref}}$
CV-CO	$A_{ia,jv} = \sqrt{\frac{2S+1}{2S}}\delta_{ij}(f_{av}^0 - f_{av}^z) + \sqrt{\frac{2S+1}{2S}}K_{ai,vj}^{\text{Ref}}$
CV-CV	$A_{ia,jb} = \delta_{ij}(f_{ab}^0 - f_{ab}^z) - \delta_{ab}(f_{ji}^0 + f_{ji}^z) - \frac{1}{S}\delta_{ij}f_{ab}^z - \frac{1}{S}\delta_{ab}f_{ji}^z + K_{ai,bj}^{\text{Ref}}$
CV-OO	$A_{ia,wv} = -\frac{1}{S}\sqrt{\frac{2S+1}{2S-1}}\delta_{vw}f_{ai}^z + \sqrt{\frac{2S+1}{2S-1}}K_{ai,vw}^{\text{Ref}}$
CV-OV	$A_{ia,vb} = -\sqrt{\frac{2S+1}{2S}}\delta_{ab}(f_{vi}^0 + f_{vi}^z) + \sqrt{\frac{2S+1}{2S}}K_{ai,bv}^{\text{Ref}}$
OO-CO	$A_{ut,jv} = -\sqrt{\frac{2S}{2S-1}}\delta_{vt}(f_{ju}^0 + f_{ju}^z) + \frac{1}{\sqrt{2S(2S-1)}}\delta_{ut}(f_{jv}^0 - f_{jv}^z) + \sqrt{\frac{2S}{2S-1}}K_{tu,vj}^{\text{Ref}}$
OO-CV	$A_{ut,jb} = -\frac{1}{S}\sqrt{\frac{2S+1}{2S-1}}\delta_{ut}f_{jb}^z + \sqrt{\frac{2S+1}{2S-1}}K_{tu,bj}^{\text{Ref}}$
OO-OO	$A_{ut,wv} = \delta_{wu}(f_{tv}^0 - f_{tv}^z) - \delta_{tv}(f_{wu}^0 + f_{wu}^z) + K_{tu,vw}^{\text{Ref}}$
OO-OV	$A_{ut,vb} = \sqrt{\frac{2S}{2S-1}}\delta_{uv}(f_{tb}^0 - f_{tb}^z) - \frac{1}{\sqrt{2S(2S-1)}}\delta_{tu}(f_{vb}^0 + f_{vb}^z) + \sqrt{\frac{2S}{2S-1}}K_{tu,bv}^{\text{Ref}}$
OV-CO	$A_{ua,jv} = \frac{2S}{2S-1}K_{au,vj}^{\text{Ref}} - \frac{1}{2S-1}K_{av,uj}^{\text{Ref}}$
OV-CV	$A_{ua,jb} = -\sqrt{\frac{2S+1}{2S}}\delta_{ab}(f_{ju}^0 + f_{ju}^z) + \sqrt{\frac{2S+1}{2S}}K_{au,bj}^{\text{Ref}}$
OV-OO	$A_{ua,wv} = \sqrt{\frac{2S}{2S-1}}\delta_{uw}(f_{av}^0 - f_{av}^z) - \frac{1}{\sqrt{2S(2S-1)}}\delta_{vw}(f_{au}^0 + f_{au}^z) + \sqrt{\frac{2S}{2S-1}}K_{au,vw}^{\text{Ref}}$
OV-OV	$A_{ua,vb} = \delta_{uv}(f_{ab}^0 - f_{ab}^z) - \delta_{ab}(f_{vu}^0 + f_{vu}^z) - \frac{2}{2S-1}\delta_{uv}f_{ab}^z + K_{au,bv}^{\text{Ref}} + \frac{1}{2S-1}K_{ab,uv}^{\text{Ref}}$